

Exchange Rate Depreciation and Inflation in Bangladesh: Evidence from Monthly Economic Data

Inflation and exchange-rate movements are important macroeconomic issues for Bangladesh. This study examines whether exchange-rate depreciation is associated with higher inflation and whether such an association extends beyond the contemporaneous month. Monthly data from the World Bank Global Economic Monitor (GEM) covering July 2006 to June 2026 are used. The analysis considers year-on-year inflation and monthly changes in the official exchange rate, together with one-month and two-month lags. Ordinary least squares regression is used to examine the relationship, while diagnostic tests and Newey–West standard errors are used to address serial correlation in the residuals. The results indicate a positive statistical association between exchange-rate changes and inflation in the current month and in the following two months. The final model has an R^2 of 0.0808, indicating that the exchange-rate variables explain a relatively small share of total inflation variation. The findings provide evidence of an association between exchange-rate movements and inflation but do not establish a causal relationship.

1. Introduction

Inflation is an important economic issue because sustained increases in the general price level reduce the purchasing power of households and can create challenges for macroeconomic stability. Exchange-rate movements may also affect domestic prices, particularly when changes in the exchange rate alter the domestic-currency cost of imported goods and production inputs.

For Bangladesh, this relationship is particularly relevant because the economy is exposed to international prices through imports and other external economic channels. A depreciation of the taka can increase the domestic-currency cost of imported goods and inputs, potentially creating upward pressure on domestic prices.

The main research question of this study is:

Is exchange-rate depreciation associated with higher inflation in Bangladesh?

The study also examines whether exchange-rate changes occurring one and two months earlier are associated with current inflation. The purpose is not to construct a complete model of inflation determination, but to provide a simple and transparent empirical investigation using monthly data.

The analysis uses World Bank GEM data for Bangladesh from July 2006 to June 2026. The study combines data preparation, visualization, correlation analysis, OLS regression, lagged regression, and diagnostic procedures in R.

2. Economic Background

Exchange-rate pass-through refers to the extent to which changes in the exchange rate are reflected in domestic prices. When the domestic currency depreciates, imported goods and imported production inputs become more expensive in local-currency terms. Firms may respond by increasing the prices of goods and services that use these imported inputs. The effect does not necessarily occur immediately because firms may use existing inventories, absorb part of the increased cost, or adjust prices gradually. Therefore, exchange-rate movements in previous months may also be associated with current inflation.

This mechanism is particularly relevant for **Bangladesh because of the country's dependence on international trade and imported inputs**. Bangladesh imports a wide range of goods and materials used by households and businesses. These include fuel and energy-related products, food and agricultural inputs, industrial raw materials, machinery, and other intermediate goods. Changes in the exchange rate against major international currencies, particularly the US dollar, can therefore influence the domestic cost of imports.

The exchange rate is also important for Bangladesh's export-oriented economy. The country has a large export sector, particularly the ready-made garment industry, while many exporting firms also depend on imported raw materials, machinery, and intermediate inputs. As a result, exchange-rate movements can affect both the domestic-currency cost of imported inputs and the revenues of exporters when foreign earnings are converted into taka.

The relationship between exchange rates and inflation became especially relevant during periods of pressure on Bangladesh's external sector. A depreciation of the taka can increase the local-currency price of imported commodities and inputs, which may raise production and transportation costs and eventually contribute to consumer price increases. At the same time, the effect of depreciation on inflation is not automatic or uniform because domestic prices are also affected by factors such as food supply conditions, energy prices, domestic demand, production costs, and monetary conditions.

These features provide the economic motivation for examining exchange-rate pass-through in Bangladesh. This study therefore investigates whether changes in the exchange rate are statistically associated with inflation during **July 2006–June 2026**. In addition to the contemporaneous relationship, the analysis examines one- and two-month lags to capture the possibility that exchange-rate changes take time to affect domestic prices. The analysis identifies statistical associations and does not assume that exchange-rate depreciation is the sole cause of inflation.

3. Data and Methodology

3.1 Data and Variables

The study uses monthly data from the World Bank Global Economic Monitor (GEM) for Bangladesh. The sample covers **July 2006 to June 2026**, giving 240 monthly observations.

The original dataset contains five main economic series:

- CPI Price;
- CPI Price, percentage year-on-year;
- official exchange rate, local currency units per US dollar, period average;
- exports of merchandise; and
- imports of merchandise.

For the empirical analysis, the main dependent variable is **inflation**, measured as the year-on-year percentage change in the CPI. This measure compares the CPI in a given month with the CPI in the corresponding month of the previous year.

The main explanatory variable is the monthly percentage change in the official exchange rate:

$$\text{Exchange Rate Change (\%)} = [(\text{Exchange Rate}_t / \text{Exchange Rate}_{(t-1)}) - 1] \times 100$$

Because the exchange rate is measured in local currency units per US dollar, an increase in this measure represents a depreciation of the domestic currency relative to the US dollar.

The dataset also contains exports, imports, trade balance ($\text{Trade Balance}_t = \text{Exports}_t - \text{Imports}_t$), export growth ($\text{Export Growth (\%)} = [(\text{Exports}_t / \text{Exports}_{(t-1)}) - 1] \times 100$), and import growth ($\text{Import Growth (\%)} = [(\text{Imports}_t / \text{Imports}_{(t-1)}) - 1] \times 100$).

These variables were prepared during the data-processing stage but were not included in the final regression model.

3.2 Data Preparation

The original GEM dataset was provided in a wide format, with the economic series represented as rows and monthly observations represented across columns. The data were first imported into R and inspected to identify the structure of the dataset and the available monthly observations.

The World Bank uses .. to indicate missing observations in the downloaded data. These entries were converted to NA values during the cleaning process rather than being treated as numerical values. The missing trade observations occurred toward the end of the sample and were retained as missing rather than being estimated or artificially filled.

The data were then reshaped into a long format so that each observation could be identified by series and month. Monthly date labels were cleaned and converted into R date values. The dataset was subsequently transformed into a one-row-per-month analysis dataset.

The final data analysis dataset contains 240 monthly observations and the following variables:

- Date
- CPI
- Inflation
- Exchange_Rate
- Exports
- Imports
- Exchange_Rate_Change
- Trade_Balance
- Export_Growth
- Import_Growth

The exchange-rate-change variable has one missing observation at the beginning of the sample because calculating a percentage change requires the previous month's exchange rate. This is expected and does not represent a data-quality problem.

Lagged variables were then created using the exchange-rate-change series. Adding the one-month and two-month lags reduces the estimation sample because observations at the beginning of the sample do not have the necessary preceding months. The final two-month lag model therefore uses 237 observations.

3.3 Model Specification

The empirical analysis begins with a simple ordinary least squares regression:

$$\text{Inflation}_t = \beta_0 + \beta_1 \text{Exchange Rate Change}_t + \varepsilon_t$$

This baseline model examines whether current-month exchange-rate changes are statistically associated with current inflation.

The analysis is then extended by including a one-month lag:

$$\begin{aligned} \text{Inflation}_t = & \beta_0 + \beta_1 \text{Exchange Rate Change}_t \\ & + \beta_2 \text{Exchange Rate Change}_{(t-1)} + \varepsilon_t \end{aligned}$$

Finally, a two-month lag is added:

$$\begin{aligned} \text{Inflation}_t = & \beta_0 + \beta_1 \text{Exchange Rate Change}_t \\ & + \beta_2 \text{Exchange Rate Change}_{(t-1)} \\ & + \beta_3 \text{Exchange Rate Change}_{(t-2)} + \varepsilon_t \end{aligned}$$

The final model therefore examines the association between inflation and exchange-rate changes occurring in the current month, one month earlier, and two months earlier, while estimating the three exchange-rate terms simultaneously.

The models are compared to examine whether adding lagged exchange-rate variables provides a statistically significant improvement in model fit. This provides a simple way to determine whether the lagged terms contribute information beyond the contemporaneous exchange-rate change.

3.4 Diagnostic Tests and Newey–West Standard Errors

Regression diagnostics are important because conventional OLS inference relies on assumptions concerning the behavior of the residuals.

The analysis therefore examined residual plots and conducted several diagnostic procedures. The Breusch–Pagan test was used to investigate heteroskedasticity, while the Durbin–Watson test and residual autocorrelation analysis were used to examine serial correlation. A Q-Q plot was also used to assess the distributional pattern of the residuals.

The Breusch–Pagan test did not provide evidence of statistically significant heteroskedasticity, with a p-value of **0.8211**. However, the Durbin–Watson test indicated strong positive serial correlation, with a statistic of approximately **0.139** and a p-value below 0.001.

Because serial correlation can make conventional OLS standard errors and associated significance tests unreliable, the final model uses **Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard errors**.

The Newey–West correction does not change the estimated regression coefficients or R^2 . Instead, it adjusts the estimated standard errors and therefore the associated test statistics and p-values to make statistical inference more robust to serial correlation and heteroskedasticity.

This distinction is important for interpreting the results. The correction does not make the model causal and does not remove the underlying serial correlation. It provides a more appropriate basis for statistical inference given the diagnostic results.

4. Empirical Results

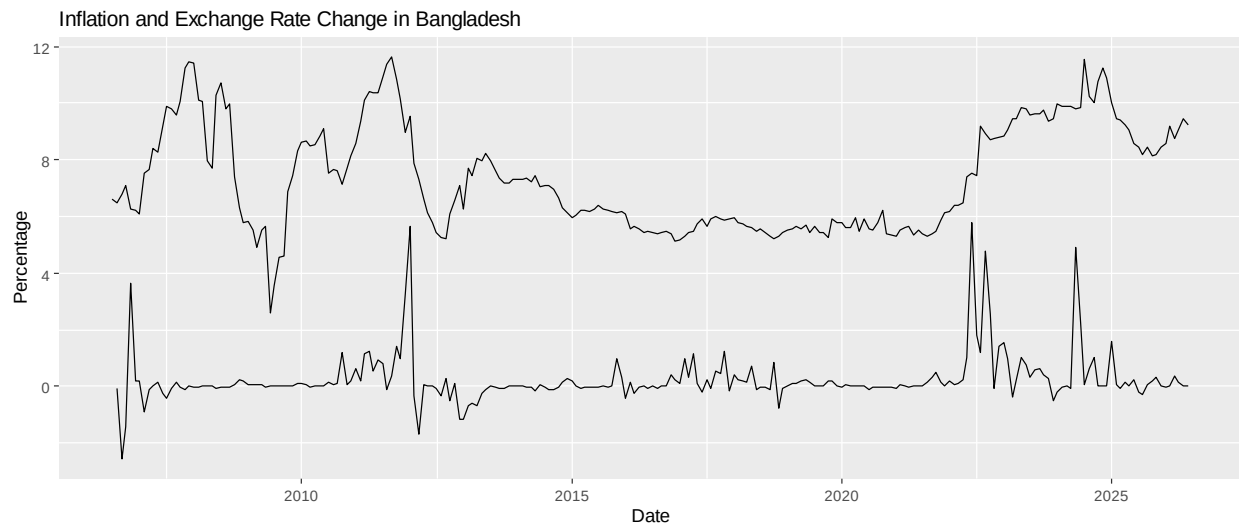
4.1 Descriptive Patterns and Visualizations

The analysis first examined the behavior of inflation and exchange-rate changes over time using graphical methods.

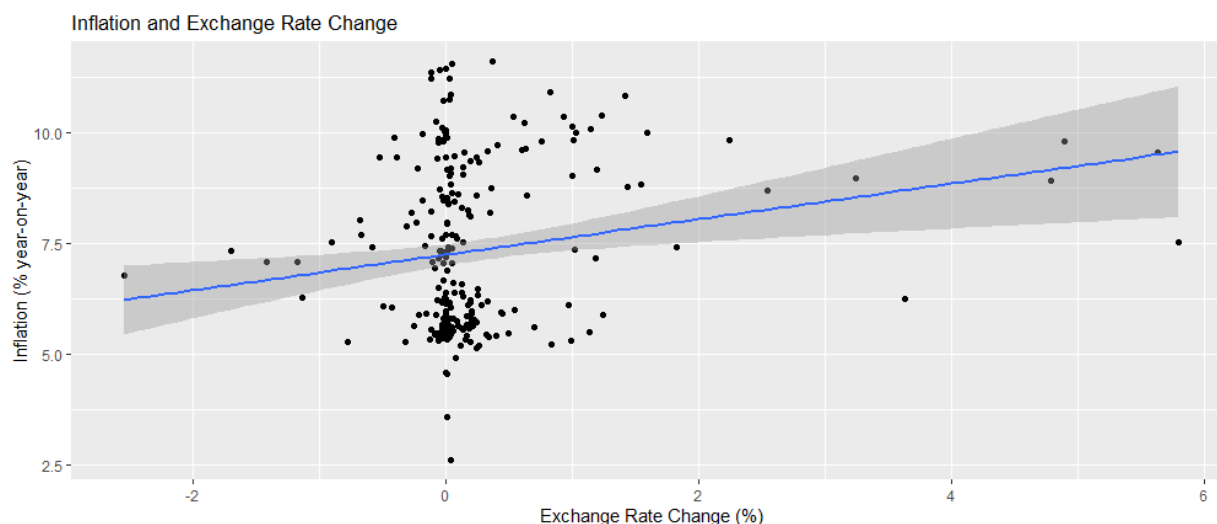
The monthly inflation series shows that inflation varies considerably over the sample period rather than remaining at a constant level. The period also contains episodes of elevated inflation, particularly during the more recent years of the sample.

The exchange-rate-change series similarly shows periods of depreciation and appreciation. Because the exchange-rate variable is expressed as local currency units per US dollar, positive percentage changes represent depreciation, while negative changes represent appreciation.

A combined time-series visualization was used to compare inflation and exchange-rate changes. The purpose of this graph was exploratory rather than to establish a formal relationship. Visual comparison can identify periods in which movements in the two series appear to coincide, but it cannot determine whether one variable causes the other.



A scatter plot was also produced with inflation on the vertical axis and exchange-rate change on the horizontal axis. The fitted regression line slopes upward, providing a visual indication of the positive association subsequently estimated in the regression analysis.



These graphical results provide an initial basis for the statistical analysis but should be interpreted together with the correlation and regression results.

4.2 Correlation Analysis

The correlation between current monthly exchange-rate changes and inflation was approximately:

$$r = 0.1913$$

This represents a **positive but relatively weak linear association** between the two variables.

The relatively small correlation is important because it indicates that exchange-rate changes and inflation do not move together in a strong one-to-one pattern. Nevertheless, a weak correlation does not necessarily mean that the exchange-rate variable has no statistically detectable relationship with inflation. Regression analysis can provide additional information, particularly when lagged terms are considered.

The correlation between inflation and the one-month lagged exchange-rate change was approximately:

$$r = 0.1920$$

The similarity between the contemporaneous and one-month lagged correlations provides some initial evidence that the association is not restricted entirely to the current month.

However, correlation is only a measure of linear association. It does not control for other variables, identify a direction of causality, or establish a transmission mechanism.

4.3 Baseline Regression

The first OLS regression examined the relationship between inflation and the contemporaneous exchange-rate change.

The estimated coefficient on exchange-rate change was approximately **0.402**, with a p-value of approximately **0.003**. The coefficient was therefore statistically significant at the 1% level.

The coefficient can be interpreted as follows: within this simple statistical model, a one-percentage-point increase in the monthly exchange-rate change is associated with approximately a **0.40 percentage-point increase in inflation**, on average.

However, this interpretation should not be presented as a causal effect.

The baseline model has an R^2 of **0.0366**, meaning that current exchange-rate changes explain approximately **3.7% of the variation in inflation** in the sample.

This relatively low R^2 is economically important. Although the estimated relationship is statistically significant, most of the variation in inflation is not explained by contemporaneous exchange-rate changes alone. Inflation is affected by numerous other economic factors that are not included in this simple model.

The baseline regression therefore provides initial evidence of a positive statistical association, while also demonstrating why a single-variable inflation model is insufficient to explain overall inflation dynamics.

4.4 Lagged Regression

The analysis was then extended to examine whether exchange-rate changes in previous months were associated with current inflation.

In the one-month lag model, both the contemporaneous exchange-rate change and the one-month lag were statistically significant. The current exchange-rate coefficient was approximately **0.291**, while the one-month lag coefficient was approximately **0.295**. The model's R^2 increased to **0.0535**.

A comparison of the contemporaneous-only and one-month lag models produced an F-statistic of approximately **4.24**, with a p-value of **0.0406**. This indicates that adding the one-month lag produced a statistically significant improvement in model fit.

The analysis was subsequently extended to two months. The final model produced the following estimated coefficients:

Variable	Coefficient
Current exchange-rate change	0.324
1-month lag	0.145
2-month lag	0.376

The model had an R^2 of **0.0808** and an adjusted R^2 of **0.0690**, compared with an R^2 of 0.0535 in the one-month lag model.

Adding the two-month lag also improved model fit, with an F-statistic of approximately **6.99** and a p-value of **0.0088**.

Because the final model exhibited strong serial correlation, the final inference relies on Newey–West standard errors. After applying the correction, all three exchange-rate terms remained statistically significant:

- Current exchange-rate change: **0.324**, $p = 0.0086$
- One-month lag: **0.145**, $p = 0.0342$
- Two-month lag: **0.376**, $p = 0.0065$

Thus, all three terms remain statistically significant at the 5% level after accounting for serial correlation.

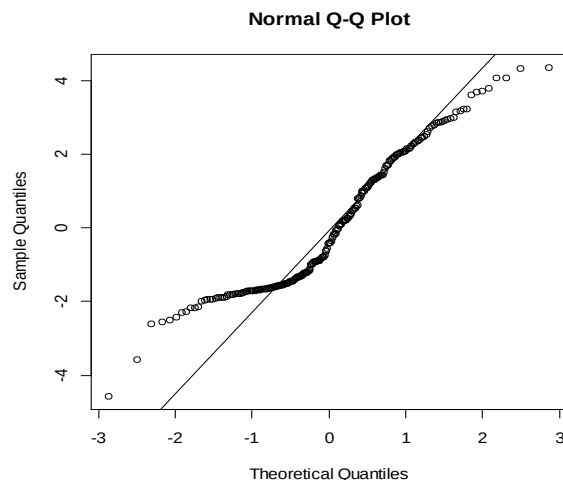
4.5 Diagnostic Results

The diagnostic analysis provides an important qualification to the regression results.

The Breusch–Pagan test did not indicate statistically significant heteroskedasticity. However, the Durbin–Watson statistic indicated strong positive serial correlation in the residuals.

Residual autocorrelation means that errors from nearby months are related rather than behaving as completely independent observations. This is plausible in monthly macroeconomic data because economic conditions and price movements often persist over time.

The Q-Q plot and residual plots were also examined as part of the diagnostic process. These procedures were used to assess whether the model displayed obvious problems in the residual behavior.



The presence of serial correlation does not automatically invalidate the estimated coefficients, but it affects the reliability of conventional standard errors. For this reason, the final statistical inference was based on Newey–West HAC standard errors.

The diagnostic results therefore influenced how the final model was reported rather than leading to the abandonment of the analysis.

4.6 Final Model and Interpretation

The final two-month lag model is statistically significant overall. Its F-statistic was approximately **6.83** with three explanatory variables, with a p-value below 0.001.

The model's R^2 of **0.0808** means that approximately 8.1% of the variation in inflation is statistically associated with the current, one-month-lagged, and two-month-lagged exchange-rate changes.

A Newey–West-adjusted joint test also indicated that the three exchange-rate variables were jointly statistically significant, with an F-statistic of approximately **3.90** and a p-value of **0.0096**.

The main empirical finding is therefore that the positive association between exchange-rate changes and inflation is not limited to the contemporaneous month. The estimated relationship remains statistically significant when exchange-rate changes from one and two months earlier are included.

The two-month lag has the largest individual coefficient, at **0.376**. However, this should not be interpreted to mean that the two-month lag alone causes inflation to increase two months later. Each coefficient is estimated conditional on the other exchange-rate terms included in the model.

The findings are therefore best summarized as evidence of a **positive statistical association between exchange-rate depreciation and inflation extending across the current month and subsequent two months**.

5. Discussion

The results indicate a positive statistical association between exchange-rate depreciation and inflation in Bangladesh. The simple regression shows that a one-percentage-point increase in monthly exchange-rate change is associated with approximately a 0.40 percentage-point increase in inflation. However, the low R^2 of

about 3.7% indicates that exchange-rate movements alone explain only a small share of changes in inflation.

The lagged analysis provides some evidence that the relationship may extend beyond the current month. In the final model, the coefficients for the current month, one-month lag, and two-month lag are all positive and statistically significant when using Newey-West standard errors. This is consistent with the idea of exchange-rate pass-through occurring gradually rather than entirely within the same month. The two-month lag has the largest estimated coefficient, although this should not be interpreted as evidence that exchange-rate depreciation causes inflation specifically after two months.

The results should therefore be interpreted as evidence of **association rather than causation**. Bangladesh's inflation is influenced by many other factors, including food prices, energy costs, imported input prices, domestic demand, and monetary conditions. These factors are not fully included in the simple regression model.

Overall, the findings support the relevance of exchange-rate movements when examining inflation in Bangladesh, while also showing that exchange-rate changes are only one part of a broader inflation process.

6. Limitations

Several limitations should be considered when interpreting the results.

First, the analysis identifies statistical associations rather than causal effects. Other economic factors may influence both exchange-rate movements and inflation, creating potential omitted-variable concerns.

Second, the residual diagnostics revealed strong positive serial correlation. Although Newey–West standard errors provide more robust statistical inference, they do not remove the underlying serial dependence.

Third, the final model has relatively low explanatory power, with an R^2 of 0.0808. Most variation in inflation therefore remains unexplained by the exchange-rate variables included in the model.

Fourth, inflation is measured as a year-on-year rate. Therefore, the dependent variable represents price changes relative to the same month of the previous year rather than a simple month-to-month inflation measure.

Fifth, the lag analysis is limited to the current month and the following two months. Exchange-rate pass-through may occur over a longer period, but longer lags are outside the scope of this introductory study.

Finally, the model does not include a broader set of macroeconomic controls. The study should therefore be viewed as a focused investigation of the exchange-rate/inflation relationship rather than a complete model of inflation determination in Bangladesh.

7. Conclusion

This study examined whether exchange-rate depreciation is associated with higher inflation in Bangladesh using monthly data from July 2006 to June 2026.

The empirical results show a positive association between exchange-rate changes and inflation. The relationship is present in the contemporaneous period and remains statistically significant for exchange-rate changes occurring one and two months earlier.

The final model has an R^2 of 0.0808, indicating that exchange-rate variables explain only a limited share of inflation variation. The use of Newey–West standard errors accounts for the strong serial correlation identified in the residuals when conducting statistical inference.

The findings are consistent with the economic possibility of exchange-rate pass-through into domestic prices. Nevertheless, the analysis does not establish causality or identify the precise mechanisms through which exchange-rate changes affect inflation.

The main conclusion is therefore that **exchange-rate depreciation is positively associated with inflation in Bangladesh in the sample studied, with the statistical association extending across the current month and subsequent two months**. Further research using additional macroeconomic variables and alternative lag structures could provide a more comprehensive understanding of Bangladesh's inflation dynamics.

8. References

World Bank. *Global Economic Monitor (GEM)*. Monthly economic data for Bangladesh.

International Monetary Fund. Analysis of inflation and exchange-rate developments in Bangladesh.

Hossain, A. (2002). Study of exchange-rate devaluation and inflation in Bangladesh.

9. Appendix

Appendix A: Main Regression Results

Variable	Baseline OLS	1-Month Model	Lag 2-Month Model	Lag
Intercept	7.248***	7.142***	—	
Current exchange-rate change	0.402***	0.291*	0.324***	
1-month lag	—	0.295*	0.145**	
2-month lag	—	—	0.376***	
R ²	0.0366	0.0535	0.0808	
Adjusted R ²	0.0325	0.0454	0.0690	
Observations	239	238	237	

Notes: The final model uses Newey–West HAC standard errors.
*** p < 0.01; ** p < 0.05; * p < 0.10.

Appendix B: Main Analytical Workflow

The analysis followed the sequence:

Raw GEM data → data cleaning → reshaping → variable construction → visualization → correlation → baseline OLS → lagged OLS → diagnostic testing → Newey–West inference → economic interpretation.

The analysis was conducted in RGui using R 4.3.1 and tidyverse tools.